

Week 8 Discussion: Options Basics & No-Arbitrage Properties

Main focus (25min): (i) intrinsic vs. time value; (ii) no-arbitrage bounds; (iii) put-call parity and early exercise intuition.

Module A) Intrinsic Value (IV) & Time Value (TV)

Option Price Components: $\text{Option Price}_t = \text{IV}_t + \text{TV}_t$

- **Intrinsic Value (IV):** The payoff if you were forced to exercise *right now*.

$$\text{IV}_t^{\text{call}} = (S_t - K)^+, \quad \text{IV}_t^{\text{put}} = (K - S_t)^+.$$

- **Time Value (TV):** The “hope” or “insurance” premium. It is the extra amount the market charges for the possibility that the stock will move favorably before expiration. $\text{TV}_t \geq 0$, and $\text{TV}_T = 0$ at maturity.

Intuition: Exercising an option early captures only the IV but instantly destroys the remaining TV.

Practice A (Textbook 10.11): Intrinsic value as a lower bound (American)

Explain why an American option is always worth at least as much as its intrinsic value.

TA Key for Practice A

At time t , the holder can always choose to exercise immediately and receive the intrinsic value IV_t . Since this exercise policy is available, the American option price must satisfy

$$V_t^{\text{Am}} \geq \text{IV}_t,$$

otherwise one could buy the underpriced option and exercise immediately for an arbitrage gain.

Module B) Arbitrage-Free Bounds (Non-dividend stock)

- **Upper Bounds:** $C_t \leq S_t$ and $P_t \leq Ke^{-r(T-t)}$.

Intuition: A right to buy the stock can't cost more than the stock itself. A right to sell for K can't be worth more than the present value of the maximum possible payoff (K).

- **Lower Bounds:**

$$C_t \geq (S_t - Ke^{-r(T-t)})^+ \quad \text{and} \quad P_t \geq (Ke^{-r(T-t)} - S_t)^+$$

Intuition (Call Lower Bound): Imagine two portfolios:

Portfolio A: Buy Call + Save PV of K in the bank. (Worth $C_t + Ke^{-r(T-t)}$)

Portfolio B: Buy the Stock. (Worth S_t)

At maturity, Portfolio A *always* does at least as well as Portfolio B. Therefore, it must cost at least as much today.

Practice B (Textbook 11.1): Lower bound for a call

What is a lower bound for the price of a 4-month call option on a non-dividend-paying stock when $S_0 = \$28$, $K = \$25$, and $r = 8\%$ per annum?

TA Key for Practice B

Use $C_0 \geq (S_0 - Ke^{-rT})^+$ with $T = 4/12 = 1/3$:

$$C_0 \geq 28 - 25e^{-0.08/3}.$$

Numerically $e^{-0.026666...} \approx 0.9737$, so $25e^{-0.08/3} \approx 24.34$, hence $C_0 \geq 28 - 24.34 = 3.66$. The lower bound is \$3.66.

Module C) Put–Call Parity & Early Exercise (Non-dividend stock)**Put–Call Parity (European):**

$$C_t(K, T) - P_t(K, T) = S_t - Ke^{-r(T-t)}.$$

Intuition (Synthetic Stock): Buying a call and selling a put exactly replicates the payoff of buying the stock using borrowed money (Forward contract).

American Call (no dividends): $C_t = C_t$. It is *never* optimal to exercise early. Why?

1. **Time Value of Money (TVM):** Why pay the strike price K today when you can keep the cash in the bank, earn interest, and pay K at maturity?
2. **Killing the Insurance:** Exercising kills the remaining Time Value (TV). If you want out of the position, you should *sell* the call in the market to capture (IV + TV) rather than exercising to only get IV.

Practice C1 (Textbook 11.6): Put–call parity computation

The price of a non-dividend-paying stock is \$19 and the price of a 3-month European call option on the stock with a strike price of \$20 is \$1. The risk-free rate is 4% per annum. What is the price of a 3-month European put option with a strike price of \$20?

Practice C2 (Textbook 11.3): Early exercise (conceptual)

Give **two** reasons why the early exercise of an American call option on a non-dividend-paying stock is not optimal. The first reason should involve the **time value of money**. The second should apply even if **interest rates are zero**.

TA Key for Practice C1–C2

C1. With $T = 3/12 = 0.25$:

$$C_0 - P_0 = S_0 - Ke^{-rT} \Rightarrow P_0 = C_0 - S_0 + Ke^{-rT} = 1 - 19 + 20e^{-0.04 \cdot 0.25} \approx 1.80.$$

C2.

- **(TVM):** Exercising early means paying the strike price K now instead of at time T . This costs the option holder the interest that could have been earned on K during the remaining life of the option.
- **(Even if $r = 0$):** Exercising destroys optionality. The call option typically has a positive Time Value ($TV > 0$) because the stock could still go up or down. Selling the call in the open market captures both Intrinsic Value and Time Value, which dominates exercising to only capture Intrinsic Value.